

Advanced Independent Torque Vectoring & Stability Architecture

For EV 3-Wheelers

Team Analog: Arpit Patel & Athul

Bajaj Engineering Pitch

Executive Summary:

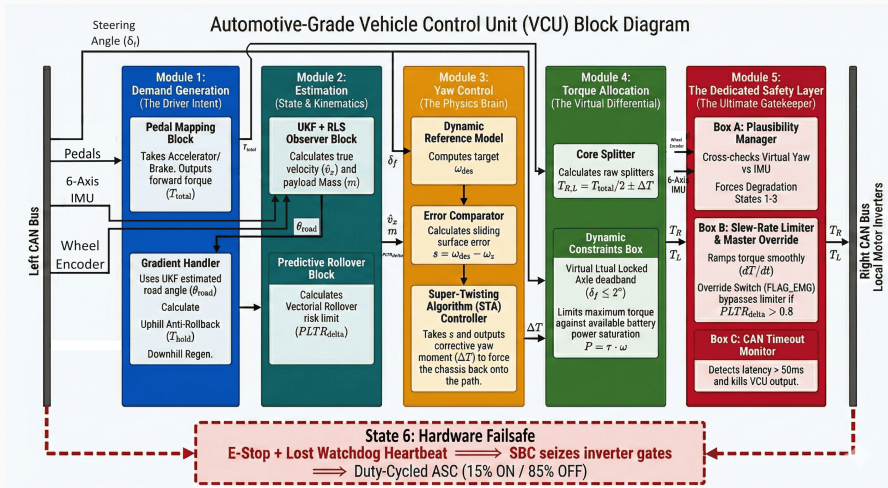
A software-defined Vehicle Control Unit (VCU) that transforms independent rear hub motors into a smart electronic differential—guaranteeing automotive-grade safety, actively mitigating triangular-footprint rollover, and preserving a natural driving feel under commercial sensor constraints.

System Architecture & Sensor Suite

- **Hardware Foundation: EV-Optimized Execution**
 - **Dual Hub Motors:** Replaces differential, enabling torque vectoring (ΔT) & regen.
 - **Un-Driven Front:** Provides baseline v_{front} ; highly vulnerable to emergency lockup.
- **Sensor Suite, Physical Inputs & Hardware Failsafes**
 - **Driver Controls:** SAS (δ_f), Pedals (demand), and hardwired **E-Stop** (Hardware override).
 - **6-Axis IMU & Encoders:** Captures ω_z , a_x , a_y , and raw tire slip (λ).
- **State Estimation: UKF + RLS Architecture**
 - **MCU-Optimized Estimation:** UKF tracks non-linear states without heavy Jacobian math, while Recursive Least Square (RLS) provides robust, real-time mass estimation.
 - **Signal Conditioning:** LPF smooths raw encoder micro-stutters before UKF fusion.
 - **Slip-Compensated Pseudo-Velocity ($\hat{v}_x$):** During front lockup ($< -1.2g$), UKF drops corrupt front data ($W_f = 0$), rejects IMU bias ($\hat{b}_x$), and blends slip-compensated rear speeds:

$$\hat{v}_x = \int (a_x - \hat{b}_x) dt + W_f v_{front} + W_r \left[\frac{(1 - \lambda_{RL})v_{RL} + (1 - \lambda_{RR})v_{RR}}{2} \right]$$

Deep-Level VCU Block Diagram Architecture



The Reference Model, Control Error & Yaw Control

- **The Dynamic Reference Model (Driver Intent)**
 - **Function:** Translates physical inputs to calculate ideal rotation rate via a 1F2R tricycle model.
 - **Target Yaw Rate:** $\omega_{des} = \frac{v_x \cdot \delta_f}{L + K'_{us} \cdot v_x^2}$
(L : wheelbase; K'_{us} : dynamic understeer gradient factoring track width B & feedforward torque M_z).
- **Kinematic Error (Sliding Surface):** $s = \omega_{des} - \omega_z$
- **Second-Order Sliding Mode Control (SOSM)**
 - **The Controller:** The VCU utilizes the Super-Twisting Algorithm (STA) to drive the kinematic yaw error (s) precisely to zero.
 - **The Benefit:** STA delivers continuous control, eliminating motor chattering. It guarantees robust directional stability across massive payload shifts (400–900 kg) without complex gain-scheduling.
 - **Corrective Torque (ΔT):** $\Delta T = -\lambda_1 |s|^{1/2} \text{sgn}(s) - \lambda_2 \int \text{sgn}(s) dt$
(Where λ_1, λ_2 are universal tuning gains stable across all loading conditions).

Predictive Rollover & RLS Payload Estimation

- **The Delta Tipping Geometry (Correcting the 4-Wheel Fallacy)**
 - **The Reality:** 3-wheelers roll over a diagonal tipping axis. VCU calculates tipping angle $\alpha = \arctan(L/(B/2))$ and the true orthogonal lever arm (d_{tip}).
- **Payload Variance & Dynamic CG Tracking**
 - **The Challenge:** Commercial payloads vary massively (400–900 kg), drastically shifting the vertical center of gravity (h) and rendering static rollover limits obsolete.
 - **The Solution:** RLS dynamically estimates total mass from longitudinal accelerations. This feeds a Mass-to-CG LUT to instantly map true vertical CG height (h).
- **The Vectorial Rollover Equation ($PLTR_{delta}$)**
 - **The Math:** Projects combined braking (a_x) & cornering (a_y) diagonally, using steering-rate ($\dot{\delta}_f$) foresight to anticipate load transfer before physical roll:
$$PLTR_{delta} = \frac{g \cdot d_{tip}}{h} (a_y \cos \alpha + a_x \sin \alpha) + C_p (\dot{\delta}_f \cdot v_x)$$
 - **Danger Mode Intervention ($PLTR_{delta} > 0.8$):** Imminent lift predicted. VCU commands massive outer-hub regen to rapidly decelerate and safely pull the CG back inside the stability triangle.

Torque Allocator, Power Limits & Steer Mitigation

- **Torque Allocator (The “Virtual Differential”)**
 - **Core Logic:** Splits longitudinal Driver Requested Torque (T_{total}) and SOSM Corrective Yaw Torque (ΔT) across the independent rear axle. **Equation:** $T_{R,L} = \frac{T_{total}}{2} \pm \Delta T$
- **Actuator Saturation & Power Limits**
 - **Dynamic Mapping:** Limits max torque based on wheel speed and battery power ($P = \tau \cdot \omega$).
 - **Strict Priority:** Shrinks forward demand (T_{total}) if limits are exceeded; yaw (ΔT) is strictly prioritized to maintain stability.
- **Torque Steer Mitigation (Drivability vs. Survivability)**
 - **Slew Limiter (dT/dt):** Torque ramps smoothly over 200–300ms to prevent violent snap.
 - **Override (FLAG_EMG):** 0ms delay max torque for threats: **[Rollover — Yaw Spike — Slip > 15% — E-Stop]**.
- **Straight-Line Stability & Progressive Blending**
 - **Virtual Locked Axle:** Enforces $T_L = T_R$ when steering is strictly straight ($\delta_f \leq 2^\circ$).
 - **Deadband (W_{steer}):** Past 2° , scales 0.0 to 1.0 ($\Delta T_{actual} = W_{steer} \cdot \Delta T$) to unlock rear axle.

Longitudinal Dynamics & Gradient Handling

- **Real-Time Gradient Estimation (UKF Integrated)**
 - **The Function:** UKF continuously estimates true road angle (θ_{road}) as an independent state, separating gravity's pull from physical chassis suspension pitch.
 - **Pitch-Compensated Equation:** $\theta_{road} = \arcsin\left(\frac{a_{x,IMU} - \dot{v}_x}{g}\right) - K_{pitch} \cdot \hat{m} \cdot \dot{v}_x \cdot h_{LUT}$
(Where K_{pitch} is the known suspension stiffness).
- **Uphill Launch: Anti-Rollback & Pitch-Over Prevention**
 - **Hold Logic:** Calculates exact minimum torque to counteract gravity using RLS mass ($\hat{m}$) & UKF gradient ($\hat{\theta}_{road}$): $T_{hold} = \hat{m} \cdot g \cdot \sin(\hat{\theta}_{road}) \cdot r_{tire}$
 - **The Execution:** Commands $T_{command} = \max(T_{hold}, T_{driver})$. Hub motors hold the trike still at 0 km/h, blending power only when driver demand exceeds gravity.
- **Downhill Descent: Preserving “Natural Driving Feel”**
 - **0% Pedal Map:** When pointed downhill ($\theta_{road} < 0$) and the driver lifts off the throttle, VCU automatically commands regenerative braking proportional to the hill's exact pull.
 - **The Benefit:** Maintains steady descents, drastically reducing mechanical friction brake wear. 7

Condition Coverage & Key Assumptions

- **Transient Instability (The Pothole Dodge & Slalom)**
 - **The Challenge:** High-speed steering flicks cause rapid weight transfer, risking un-tripped roll.
 - **The Solution:** VCU uses derivatives. Violent maneuvers spike steering ($\dot{\delta}_f$) & yaw error ($\dot{e}_r$).
 - **The Result:** Triggers FLAG_EMG override, dumping corrective torque (ΔT) to stabilize.
- **μ -Split Surfaces & Active Slip Control ($\sim 10\text{--}15\%$ Peak Traction)**
 - **The Challenge:** Asymmetric friction induces massive torque steer if motors are unrestricted.
 - **μ -Split Protection:** UKF monitors wheel acceleration ($\dot{\omega}$). Sudden spikes dynamically inflate noise variance, driving Kalman Gain to zero to seamlessly reject slipping data.
 - **Slip Calc:** Compares hub vs. chassis speed ($\hat{v}_x$):
$$\lambda = \frac{|\omega_{wheel} \cdot r_{tire} - \hat{v}_x|}{\max(|\omega_{wheel} \cdot r_{tire}|, |\hat{v}_x|)}$$
 - **Friction Saturation Interlock:** If front slip ($\lambda_f \rightarrow 1.0$) signals grip loss, target yaw moment (M_z) is instantly attenuated to zero to prevent catastrophic spin-outs.
- **Key Assumptions & Simplifications**
 - **Parameters:** Payload mass and CG height are unknown at key-on. Suspension stiffness and track geometry are treated as rigid constants.
 - **System Model:** Employs a First-Order 1F2R Tricycle Kinematic Model. Hub motor torque response is assumed to be instantaneous relative to chassis inertia.

Fault Handling, Plausibility & Degradation States

- **The “Lie Detector” (Analytical Redundancy)**
 - **Plausibility Check:** VCU continuously cross-checks physical sensor inputs against baseline kinematic models to guarantee data integrity.
 - **Virtual Yaw Validation:** $\omega_{virtual} = \frac{V_{RL} - V_{RR}}{T_{track}}$. VCU instantly isolates IMU data if it severely diverges from this independent rear-wheel calculation.
- **The Degradation State Machine**
 - **State 1 (Steering Sensor Loss):** VCU abandons active target vectoring. Relies on IMU to apply Yaw Damping, resisting uncommanded rotation to prevent front-end plowing.
 - **State 2 (IMU Loss):** Without yaw feedback, VCU enforces severe torque limits ($< 20\%$) and strictly caps speed to initiate a 15 km/h limp-home mode.
 - **State 3 (Encoder Loss):** The Hybrid Observer seamlessly transitions to pure IMU dead-reckoning to maintain state estimation without traction data.
 - **State 4 (CAN Timeout $> 50\text{ms}$):** Local inverters detect network silence and autonomously execute a smooth 0.2g regenerative deceleration.

Critical Failsafes, Pedal Faults & Hardware Integration

- **State 5: Pedal Sensor Faults & Plausibility Conflicts**
 - **Sensor Death (Signal Loss):** Pedal potentiometer fails. **Fallback:** VCU forces “Zero-Torque Coasting”. Disables active regen to prevent sudden deceleration, requiring mechanical stop.
 - **Plausibility Conflict (Stuck Pedal):** Simultaneous accel/brake signals (e.g., jammed throttle). **Fallback:** Brake Override System (BOS) ignores accelerator when brake pressure crosses safety threshold.
- **State 6: Total System Collapse (Ultimate Hardware Failsafe)**
 - **The Nightmare:** Primary VCU freezes at max speed on steep downgrade, rendering software failsafes useless.
 - **Independent Watchdog & E-Stop:** Independent System Basis Chip (SBC) monitors VCU heartbeat. If lost AND driver hits E-Stop, SBC isolates VCU and seizes inverter gate drivers.
 - **Thermal-Safe ASC (Duty-Cycled):** Hardware multivibrator applies fixed PWM (15% ON / 85% OFF) to Active Short Circuit pins, shedding kinetic energy without overheating.
 - **Physics Limit & Handoff:** As vehicle slows, back-EMF drops. At ~ 15 km/h, drag tapers off; driver applies mechanical brake for a spin-free stop.